Leetcode 88 :- Merge Sorted Array

$i = 0, j = 0, k = 0;$

[10, 20, 40, 80, 90]

[5, 12, 43, 77, 88]

extension

[1, 2, 3, 4]

[100, 200, 300, 400]

[5, 10,]

res = [5, 10, 12, 20, 40, 43, 77, 80, 88, 90]

```
if (nums1[i] < nums2[j]) {
    res[k++] = nums1[i++];
}
else {
    res[k++] = nums2[j++];
}
```

```
while(i < nums1.length && j < nums2.length){
  if(nums1[i] < nums2[j])
      res[k++] = nums1[i++]
    else res[k++] = nums2[j++]
}
```

k = 0 1

[1, 2, 3, 5] [4, 100, 200, 300, 400,]

[1, 2, 3, 4, 5,

// for remaining elements

```
while(i < nums1.length){
    res[k++] = nums1[i++];
}
while(j < nums2.length){
    res[k++] = nums2[j++];
}
```

```
              n1                   n2
int i = nums1.length - 1, j = nums2.length - 1;
int k = n1+n2-1;
while(i>=0 && j>=0) {
   if(nums1[i] > nums2[j])
        res[k--] = nums1[i--];
   else  res[k--] = nums2[j--];
}
```

```
while(i>=0)
   res[k--] = nums1[i--];
while(j>=0)
   res[k--] = nums2[j--];
```

Leetcode 134 :-

gas(i) = (1, 2, 3, 4, 5)

0 1 2 3 4

cost(i) = [3, 4, 5, 1, 2]

0 1 2 3 4

gas(i) - cost(i)

(1, 2, 3, 4, 5) (1, 2, 3, 4, 5)

4 - 1 = 3 + 5 = 8 - 2 = 6 + 1 = 7 - 3 = 4

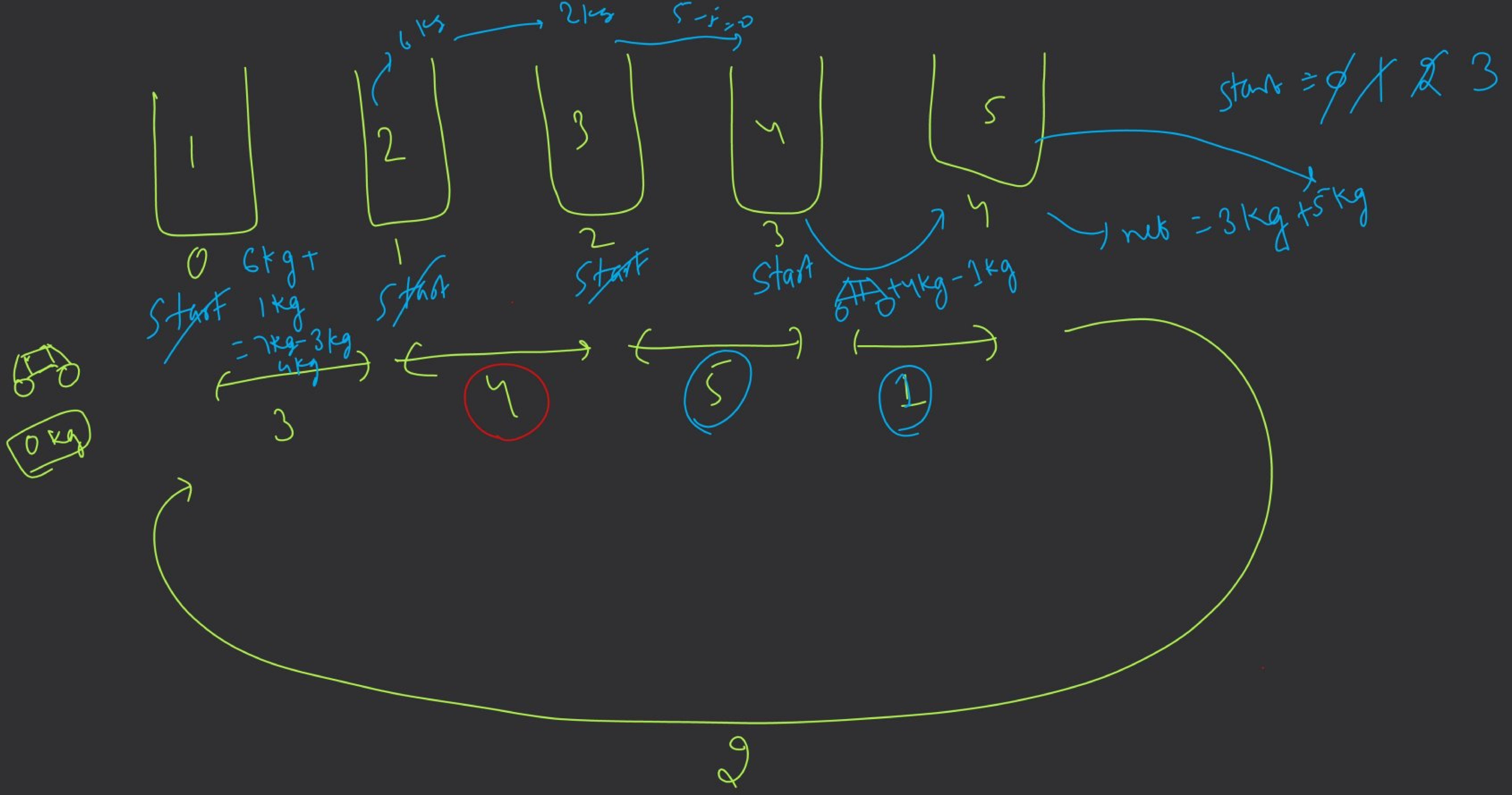
4 + 4 = 8 - 2 = 6

6 + 5 = 11 - 3 = 8

→ Total gas >= Total cost

→ false

→ (return -1)



1
2
3
4
5
0
1
2
3
4
Start
Start
Start
Start
6kg
2kg
0 kg
3
4
5
1
9
net = 3kg + 5kg

Gas = [2, 3, 4] = 9

Cost = [3, 4, 3] = 10

2, 1, 0 ✓

(failed here)

Gas = [1, 2, 4, 5, 9]

start st st (start)

Cost = [3, 4, 1, 10, 1]

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Description | Note | Editorial | Solutions

Discuss Approach

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You are given two integer arrays `nums1` and `nums2`, sorted in **non-decreasing order**, and two integers `m` and `n`, representing the number of elements in `nums1` and `nums2` respectively.

Merge `nums1` and `nums2` into a single array sorted in **non-decreasing order**.

The final sorted array should not be returned by the function, but instead be *stored inside the array* `nums1`. To accommodate this, `nums1` has a length of `m + n`, where the first `m` elements denote the elements that should be merged, and the last `n` elements are set to `0` and should be ignored. `nums2` has a length of `n`.

(m, (n

Example 1:

Input: nums1 = [1,2,3,0,0,0], m = 3, nums2 =

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Code

Java | Auto

```java
class Solution {
    public void merge(int[] nums1, int m, int[] nums2, int n) {

    }
}
```

(k = m+n-1)

nums2 = (2,5,6)

res = [1,2,2,3,5,6]

for(i=0 → m+n) {

nums1(i++) = res(i++)

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Test Result | Testcase

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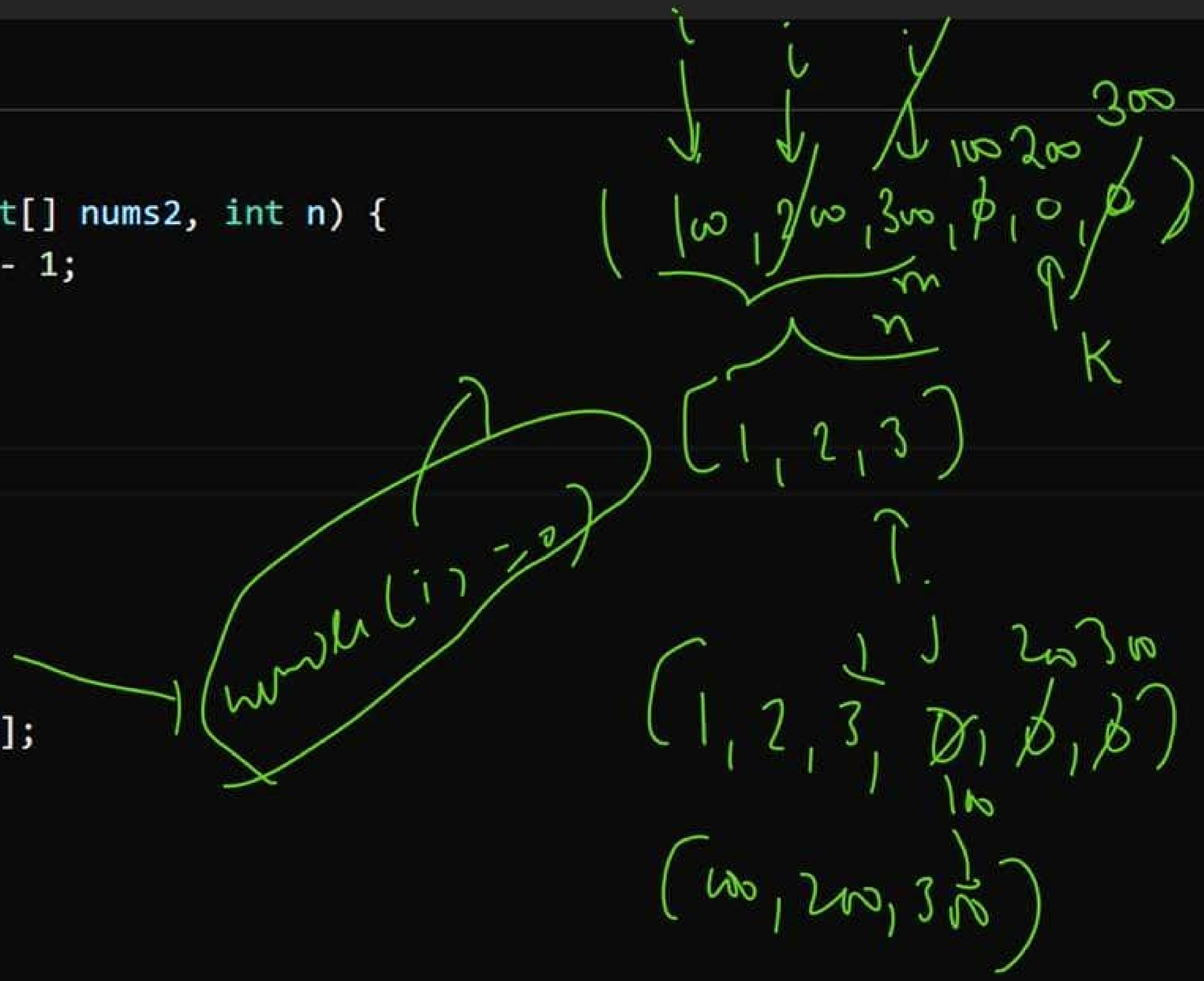
Code

Java Auto

```java
class Solution {
    public void merge(int[] nums1, int m, int[] nums2, int n) {
        int i = m - 1, j = n - 1, k = m + n - 1;

        while(i >= 0 && j >= 0){
            if(nums1[i] > nums2[j])
            nums1[k--] = nums1[i--];
            else
            nums1[k--] = nums2[j--];
        }

        while(j >= 0) nums1[k--] = nums2[j--];
    }
}
```



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Java Auto

```java
class Solution {
    public int canCompleteCircuit(int[] gas, int[] cost) {
        int totalGas = 0, totalCost = 0;
        for(int num : gas){
            totalGas += num;
        }
        for(int num : cost){
            totalCost += num;
        }
        if(totalGas < totalCost) return -1;
        int start = 0,currGas = 0;
        for(int i = 0;i<gas.length;i++){
            currGas += (gas[i] - cost[i]);

            if(currGas < 0) {
                start = i + 1;
                currGas = 0;
            }
        }
        return start;
    }
}
```

Totalgas = 21

total cost = 19

start start start start

i i i i

Gas = [1, 2, 4, 5, 9]

Cost = [3 4 1 10 1]

CurrGas = ~~-2~~ ~~0~~ ~~-2~~ ~~0~~ ~~3~~ ~~-2~~ 0

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Code

Java Auto

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class Solution {
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        int start = 0,currGas = 0;
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            currGas += (gas[i] - cost[i]);

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        }
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}
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totalGas = 21

total cost = 19

start start start i start

Gas = [1, 2, 4, 5, 9, 1]

Cost = [3 4 1 10

CurrGas = ~~-2~~ ~~0~~ ~~-2~~ ~~0~~ ~~3~~ ~~-2~~ 0